was set aside to assist the police and courts service in handling the huge volumes of CCTV images these systems produced. As a result, for a long time, the police largely turned a blind eye to the existence of any CCTV evidence, unless the case was of a very high profile. It is now understood that CCTV only works as part of an integrated programme that takes CCTV images all the way from camera to court. It is no good generating images unless some action is taken as a result – whether that is deploying police to the scene, presenting images in court, or detaining those identified driving stolen vehicles.

- *Technical standards.* Somewhat after the event, the UK is now very keen to retro-actively apply a swathe of technical standards for the type of CCTV kit that is allowed to be deployed to view members of the public. This is because CCTV from private sources – shopping centres, offices and so on – is often of insufficient quality for use in court. It has also been because the migration from analogue to digital operation across the whole of the PSS CCTV estate has caused so much difficulty. At the last count, there were over 800 different technical standards for image data storage on DVRs. This has made image recovery following any incident enormously more time consuming, sometimes even impossible. For example, some DVR systems have no facility for saving data out to disk. Others save data out, but without the necessary viewer software needed to view the images.
- *Standards for operation.* Protections for the public against being inappropriately surveilled are weak and the law for so doing lacks teeth. Also, many criminal prosecutions involving CCTV image data from privately operated CCTV systems continue to fail – usually because of mistakes by CCTV owners in carrying out basic procedures to ensure the necessary audit trail of the evidence (for example, trivial problems like having the automated date/time image watermarking system on the recorder incorrectly set).
- *Successful CCTV operation is mostly about people.* Contrary to what was initially thought, research now shows that only a few people have the ability to spot incidents seen by cameras at the monitoring centre. It turns out to be a skill that some people have and others don't. Specialist staff selection methods have had to be developed. Also, research undertaken on behalf of the airline industry – looking at security staff viewing baggage X-ray machines – shows that even those with the correct aptitude can only maintain the necessary level of concentration for short periods at a time – perhaps only for 20minutes or so. CCTV monitoring is often a mundane occupation, so there is a need to rotate staff through other duties at the monitoring centre for variation, and also to develop programmes whereby good staff can progress – otherwise staff churn will continue to be a significant problem.

NEW TECHNOLOGY

Surveilling the future

Once captured, a digital video image can now be analysed using software tools. This has lead to a number of new or emerging CCTV applications:

Video analytics: Here, AI-based software automatically analyses the video data looking for pre-identified patterns of behaviour. When such behaviour is identified, pre-set actions are then triggered. An enormous amount of research is underway in this area, since it has the potential to aid monitoring staff by highlighting potentially significant events for their scrutiny. However, the technology is still very much in its infancy. Plagued by high false positive rates, only the very simplest of these applications have so far been shown to be commercially deployable. Generally these applications require a very controlled environment to operate effectively. For example, video analytics is used to reduce employee fraud in supermarkets by reconciling shape recognition in CCTV data of items filmed on the check-out belt with EPOS data from items keyed by checkout staff. It is also used in an anti-terrorist application to detect abandoned packages on, for example, railway platforms or other public areas where people gather. Software 'learns' what the camera view looks like, and compares successive image frames looking for new items that remain stationary. London Underground has trialled this technology with some success. Other video analytic applications that are for the moment more of a work in progress, include algorithms to identify people running, fighting, falling, carrying guns or knives. There are algorithms for car crime (identifies human-sized object, moving from car to car to car in a car park) and potential suicides on railway platforms (apparently, would-be suicides stand as far away from the platform edge as possible and then move to the edge, and repeat this activity several times before making their attempt. This pattern of movement is in theory identifiable using video analytics).

Facial recognition: A branch of video analytics. Currently this can be made to work with a high degree of reliability, but only in very controllable environments, for example police interview suites, access control gates, or border control points. The latest generation of systems being used at border control points work by projecting a grid of light 'stripes' onto the individual's face and then using CCTV images of the distortion of the grid to build a 3D model of the face. Recently this technology has been shown to work to a useful level of reliability as travellers walk normally through a turnstile type 'pinch point' designed for the purpose. The still unresolved problem is the level of time consuming data crunching required for matching each new person passing through against the data of known 'wanted' targets. Currently it takes about three seconds for comparison, making the system still unworkable for high footfall urban transit applications.

Intelligent storage: Existing video storage systems automatically apply 'metadata' tagging to video image data to provide various search functionalities. The simplest of these metadata tags are date, time and camera ID number, allowing users to search CCTV data by time and date, or by the images from a particular camera. However, as a by-product of video analytics, if AI software can identify objects in CCTV image data then it can also automatically apply appropriate metadata tags to the image data. An example might be 'person running' or 'red truck'. In the future such metadata could be streamed live (which would require only small bandwidth) from the various CCTV systems around the country to a central real-time searchable database facility. However, for the moment, the lack of a standards for metadata tagging is hampering any such efforts.

A would be terrorist buying bomb materials, caught on CCTV